



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (Semester I) - June, 2025
STA1013 - Introduction to Statistics (3 Credits)

Answer **ALL** questions.

Time allowed: **TWO** hours.

Instructions

- This is a closed book examination.
- You are permitted to use a Scientific calculator during the examination.
- Use the answer booklets to write your answers and the single sheets for rough work.
- For any unacceptable behaviour, including the violation of examination regulations during the examination, official actions will be taken under Section 9 (Regulations Related to Exam Procedures) of the student handbook.

1. (a) Read the following scenario carefully and answer the questions.

A sociology researcher is studying lifestyle patterns among young adults in an urban community. A sample of 100 individuals aged 18–30 is surveyed. The data collected includes each participant's age (in years), monthly income (in dollars), highest level of education completed (high school, diploma, degree, or postgraduate), number of hours spent on social media per day, whether they are employed (yes or no), their marital status (single, married, divorced), and their satisfaction with life on a scale of 1 to 5. The researcher also records each participant's neighborhood (coded as A, B, C, or D) and whether they live alone or with others. In addition, participants were asked how many books they read in a month and whether they voted in the last election. Participants also provided their self-identified political leaning on a scale from 1 (very liberal) to 7 (very conservative).

- i. Identify all the variables mentioned in the scenario.
- ii. For each variable, determine whether it is qualitative or quantitative.
- iii. If the variable is quantitative, state whether it is discrete or continuous.
- iv. Specify the level of measurement (nominal, ordinal, interval, or ratio) for each variable.

[10 marks]

- (b) Data tabulation and visualization play a vital role in organizing, summarizing, and presenting statistical data effectively.

- i. What is the purpose of data tabulation? Clearly distinguish between simple and complex tables, providing an example of each.
- ii. Identify and describe three commonly used data visualization methods in descriptive statistics. For each, mention the type of data it best represents and highlight one key advantage.

[6 marks]

- (c) Measures of central tendency and dispersion, along with skewness, provide a detailed summary of data characteristics. A dataset has a mean of 80, a median of 72, and a standard deviation of 10.

- i. Based on these values, comment on the likely direction of skewness. Justify your answer.
- ii. Calculate Karl Pearson's coefficient of skewness.
- iii. Briefly explain what a positive, negative, and zero skewness value indicates about the shape of a distribution.

[9 marks]

[P.T.O.]

2. (a) Explain why standard deviation is considered a better measure of dispersion compared to other measures. Provide three reasons to support your answer. Also, state any limitations it may have. [8 marks]

- (b) The arithmetic mean of a certain distribution is 5. The second and third moments about the mean are 20 and 140, respectively. Calculate the third moment of the distribution about 10. [6 marks]

- (c) The table below shows the distribution of weights (in kilograms) of a group of 40 individuals:

Weight Range (kg)	Frequency
40 – 50	5
50 – 60	10
60 – 70	12
70 – 80	8
80 – 90	5

- Calculate the mean and standard deviation of the weights.
- Compute the fourth central moment about the mean and determine the kurtosis.
- Interpret the kurtosis value and comment on the shape of the distribution.

[11 marks]

3. A researcher collected data on the number of hours students studied per week (X) and their corresponding scores in a Statistics exam (Y). The data for a sample of 5 students is as follows:

Student ID	A	B	C	D	E
Hours Studied (X)	2	4	6	8	10
Exam Score (Y)	65	70	75	80	85

- (a) i. Clearly explain the correlation coefficient. [2 marks]
 ii. The correlation coefficient between two variables is denoted by r . Five different values of r , rounded to four decimal places, are given below: [2 marks]

$$r_1 = 0.0000, \quad r_2 = 0.9812, \quad r_3 = -1.0000, \quad r_4 = 0.7652, \quad r_5 = -0.7098$$

Match each of the following four scatter graphs, showing observations from different bivariate data sets, to one of the values of r given above. You should not use each given value of r more than once. Justify your answers.



Figure 1



Figure 2



Figure 3



Figure 4

- iii. Sketch a scatter graph for the remaining value of r from the list above. Justify your answer. [2 marks]

[Continue...]

[Question 3. continued...]

- (b) i. Using the data provided, calculate the Pearson correlation coefficient (r) between hours studied and exam score. Show all necessary steps. [4 marks]
 ii. Interpret the result obtained above [b(i)]. [2 marks]
- (c) Derive the formulas for the least squares estimators of the regression line $Y = \beta_0 + \beta_1 X$. [4 marks]
- (d) i. Using the sample data, calculate the estimated regression coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$. [6 marks]
 ii. Write the estimated regression equation. [1 mark]
 iii. Use the regression equation to predict the exam score for a student who studies for 7 hours. [1 mark]
 iv. Interpret the meaning of the slope and intercept in context. [1 mark]
4. A statistician is comparing price changes of selected commodities between two periods; Base Year (2020) and Current Year (2024). The data are given below:

Commodity	Price in 2020 (Rs)	Price in 2024 (Rs)	Quantity in 2020	Quantity in 2024
Rice	80	120	10	9
Sugar	100	150	5	4
Milk	60	72	8	10
Bread	40	60	6	7

- (a) i. Define what an index number is and explain its general purpose. [2 marks]
 ii. Briefly explain the difference between simple and weighted index numbers. [2 marks]
 iii. What does it mean to "deflate" a value using an index number? [2 marks]
- (b) i. Compute the Simple Price Relative Index for each commodity (using 2020 as base year). [2 marks]
 ii. Calculate the Simple Aggregate Price Index for the year 2024. [2 marks]
 iii. Interpret your answer drawn above [b(ii)]. [2 marks]
- (c) i. Calculate the Laspeyre's Price Index using 2020 as the base year. [2 marks]
 ii. Calculate the Paasche's Price Index using 2024 quantities as weights. [2 marks]
 iii. Compute the Fisher's Ideal Index and comment on the results. [3 marks]
- (d) i. Discuss one strength and one weakness each of Laspeyre's, Paasche's, and Fisher's index numbers. [3 marks]
 ii. Suppose a household income in 2024 is Rs. 150,000. Use the Fisher's Index to calculate its real income in 2020 prices (i.e., deflate it). [2 marks]
 iii. Interpret the result drawn above [d(ii)]. [1 mark]